# Edukom Group Platform Upgrade: Architecture & Execution Roadmap

This document outlines the end-to-end technical strategy and phased implementation for upgrading Edukom Group from external third-party software (TutorCruncher, Zoho, Stripe, Zoom, Google Meet) to a proprietary, unified, and resilient educational portal.

## 1. High-Level System Architecture

## 2. Phase Execution Progress & Final Status

| Phase | Core Deliverables | Cybersecurity & Safeguards | Test Coverage | Status |
|---|---|---|---|---|
| **Phase 1** | Multi-Currency Payments & Automated Tiered Payroll Engine | Paystack/Flutterwave HMAC webhooks, CSPRNG references, NUBAN KYC bank validation, CWE-1236 CSV injection sanitization | 14 Tests Passing | **COMPLETED & VERIFIED** |
| **Phase 2** | Embedded WebRTC Classrooms & Custom Portals | Double-booking prevention engine, dual-timezone display, Daily.co JWT generation, parent progress reports, QA rating engine | 15 Tests Passing | **COMPLETED & VERIFIED** |
| **Phase 3** | DRM Encrypted Video Library & WhatsApp Automations | **9 Safeguards**: Concurrent Stream Guard, Live Recording Auto-Ingestion, NDPR Notification Preferences & Quiet Hours, Watch Progress Resume, Inbound WhatsApp Webhook & Keyword Triage, Subscription Expiry Alerts, Adaptive Bitrate (ABR) HLS Data Saver, Minor Recording Consent Guard, Offline Encrypted PWA Caching | 19 Tests Passing | **COMPLETED & VERIFIED** |
| **Phase 4** | Reliable IT Infrastructure & SLAs | Multi-stage Dockerization, Nginx reverse proxy with HSTS/CSP, `/healthz` & `/readyz` probes, AES-256 automated backups, 99.9% uptime SLA, plus **5 Enterprise Safeguards**: DevSecOps CI/CD ( `pip-audit` , `gitleaks` , `trivy` ), Certbot SSL auto-renewal, Sentry APM with NDPR PII scrubbing, rotating audit logs, and zero-downtime deployment with auto-rollback | 4 Tests Passing | **COMPLETED & VERIFIED** |
| **TOTAL** | **Entire Edukom Enterprise LMS** | **Full 4-Phase End-to-End Enterprise Upgrade** | **52 / 52 Tests Passing (100% OK)** | **PRODUCTION READY** |

## 3. Subsystem Breakdown Across All 4 Phases

### Phase 1: Financial & Payroll Subsystem

- : Native checkout via Paystack and Flutterwave, accepting cards, NGN bank transfers, USSD, and diaspora currencies (USD, GBP, EUR). Real-time HMAC-SHA512 webhook signature verification.
- : Ingests TutorCruncher and Zoho CSV exports, validates bank account numbers against Nigerian Inter-Bank Settlement System (NIBSS) NUBAN APIs, sanitizes spreadsheet formulae against CWE-1236, and computes hourly compensation across 140 tiered matrix rules covering 5 global zones.

### Phase 2: Embedded Classrooms & Interactive Portals

- : Embedded Daily.co WebRTC classrooms. Tutors and students interact directly within Edukom without third-party links. Includes:
    1. Double-booking conflict engine preventing tutor session collisions.
    2. Multi-timezone dual time display (Local vs. WAT).
    3. Parent progress reports & tutor lesson logs.
    4. Worksheets & resource sharing repository.
    5. 24-hour structured cancellation policy.
    6. Quality assurance rating & low-score operational alerts.

### Phase 3: DRM Encrypted Video Library & WhatsApp Automations

- : AES-128 HLS encrypted video library with 9 enterprise safeguards:
    1. **Concurrent Stream Guard**: Prevents account sharing by terminating older sessions upon simultaneous logins.
    2. **Live Recording Auto-Ingestion**: Automatically bridges finished Daily.co classroom recordings into the encrypted library.
    3. **Parent Preferences & Quiet Hours**: Granular channel toggling and quiet hours suppression.
    4. **Student Watch Progress & Resume**: Remembers exact playback position and prompts resume.
    5. **Two-Way WhatsApp Inbound Support**: Captures parent replies, correlates to scheduled classes, and triages emergencies.
    6. **Subscription Expiry Alerts**: Triggers 3-day and 1-day renewal warnings with 1-click payment links.

7. **Adaptive Bitrate (ABR) Multi-Quality HLS**: Delivers 1080p down to 360p Data Saver mode for African mobile networks.
8. **Minor Recording Consent Guard**: Enforces NDPR/COPPA explicit parental consent before ingesting any minor's session.
9. **Encrypted Offline Caching**: PWA service worker caching encrypted chunks without exposing raw MP4 files.

## Phase 4: Reliable IT Infrastructure, Disaster Recovery & SLAs

- **Containerization**: Multi-stage running unprivileged non-root user `edukom` (UID 10001) and orchestrating Gunicorn, Celery, PostgreSQL 16, Redis 7, and Nginx.
- **Reverse Proxy Hardening**: enforcing TLS 1.3, HSTS ( `max-age=31536000` ), Content Security Policy (CSP), and strict rate-limiting on authentication and DRM key endpoints.
- **Health Probes**: Liveness ( `/healthz` ) and readiness ( `/readyz` ) in monitoring database connectivity, Redis roundtrip latency, and storage headroom (> 10%).
- **Disaster Recovery**: Automated backup () and point-in-time restore () with OpenSSL AES-256-CBC encryption and SHA-256 integrity verification.
- **SLA Commitment**: 99.9% uptime target with defined MTTA and MTTR tiers and 72-hour NDPR breach notification protocols in .
- **Enterprise DevSecOps & Operations**: Automated CI/CD security scanning pipeline (), automated Certbot SSL lifecycle, Sentry APM with custom NDPR PII scrubbing, rotating audit logs ( `RotatingFileHandler` ), and zero-downtime rolling deployments with automated rollback ().